

QAegisPralaya

Quantum-Secured Emergency Intelligence Platform

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Hybrid QML PQC Security Unified EOC

SDGs: 3 (Health) 11 (Cities) 16 (Justice)

The Real Problem

Urban Crisis Coordination Failure

India experiences more than 1,500 flood-related deaths annually and billions in infrastructure damage [National Disaster Management Authority, 2023].

Cascade Failures

Heavy Rainfall → Power Grid Failure → Water System Failure → Hospital Failure → Rescue Coordination Collapse

Key Coordination Gap

- Departments operate with siloed data
- No interdependency modelling
- Emergency commanders rely on unverified field reports
- No cryptographic integrity guarantees

Problem Statement

Core Challenge

Municipal Emergency Operations Centers cannot model how failures cascade across departments in real time.

Quantum Contribution 1 — QML

Hybrid Variational Quantum Circuits model cross-domain cascade risks before failure chains complete [Cerezo et al., 2021, Biamonte et al., 2017].

Quantum Contribution 2 — PQC

Dilithium3 (ML-DSA) digitally signs all field reports to ensure tamper-evident authenticated communications [National Institute of Standards and Technology, 2024].

Positioning

Quantum-ready architecture running on simulators today, designed to scale to real hardware later.

Existing Platforms Miss Critical Gaps

Platform	ML	Cross-Domain	PQC	Cascade
ESRI ArcGIS	Limited	✗	✗	✗
IBM PAIRS	Classical	✗	✗	✗
FEMA IPAWS	None	✗	✗	✗
Academic QML	✓	✗	✗	✗
QAegisPralaya	✓	✓	✓	✓

Key Insight

No existing platform combines:

- Cascading-domain prediction
- Cross-department modelling
- Post-quantum secure communications

Two Quantum Pillars

Pillar 1 — QML Cascade Prediction

Pipeline:

Climate Features → PCA → 5Q VQC → Risk Prediction

Outputs:

- Cascade risk score
- Propagation chain
- Timeline estimation

Pillar 2 — PQC Secure Communications

Pipeline:

EOC Key Distribution → Responder Signs Report → EOC Verifies

Built using Dilithium3 via liboqs-python [Open Quantum Safe Project, 2024].

Variational Quantum Circuit

Input Features

- Temperature
- Wind Speed
- Precipitation
- Grid Load
- Water Pressure
- Hospital Capacity
- Telecom Uptime
- Seismic Activity

VQC Architecture

AngleEmbedding → Entangling Layers → Measurement

- 5 qubits
- Adam optimizer

Data Infrastructure

Real Data Sources

- OpenWeatherMap API
- Open-Meteo API
- USGS Seismic Feed
- IMD Weather Data

Synthetic Scenario Generator

- 1 Sample climate variables
- 2 Apply cascading dependency rules
- 3 Inject probabilistic noise
- 4 Generate labelled crisis scenarios

Generated Scenarios

- Flood + Grid Failure
- Heatwave + Hospital Surge

Quantum ML

- PennyLane
- Variational Quantum Circuits
- Hybrid QML

Post-Quantum Security

- liboqs-python
- Dilithium3
- NIST ML-DSA

Backend

- FastAPI
- Uvicorn
- WebSockets

Frontend

- React
- Vite
- Tailwind
- Leaflet.js

System Architecture

Data Sources → QML Engine → PQC Verification → FastAPI Layer → React Dashboard

Data Sources

OpenWeatherMap, IMD, USGS, Synthetic Generator

Processing Layer

- PennyLane VQC
- Dilithium3 Signing
- Async Scenario Runner

Dashboard

- Incident Map
- Risk Gauge
- Alert Timeline

Unique Selling Points

- 1 Only platform combining QML + PQC
- 2 Cross-department cascading prediction
- 3 NIST-standard post-quantum cryptography
- 4 Quantum-ready while deployable today
- 5 Unified emergency command dashboard

Novelty

Integration novelty across:

- Quantum Machine Learning
- Post-Quantum Cryptography
- Infrastructure Cascade Modelling

4-Hour Build Plan

Parallel Development

- Agent 1A — Data Pipeline
- Agent 1B — QML Engine
- Agent 1C — PQC Layer
- Agent 2 — Dashboard + Integration

Stage Gates

- VQC accuracy $> 65\%$
- PQC verification $< 500\text{ms}$
- All API routes operational
- Demo scenarios automated

Demo Scenarios

Scenario 1 — Flood + Grid

Risk: 87%

Climate → Grid → Medical

Scenario 2 — Heatwave + Hospital

Risk: 91%

Climate → Water → Medical

Scenario 3 — Cyclone + Telecom

Risk: 79%

Climate → Telecom → Medical

SDG Alignment and Impact

SDG 3 — Health

Faster hospital coordination during crisis escalation.

SDG 11 — Sustainable Cities

Cross-domain infrastructure awareness and predictive response.

SDG 16 — Justice and Institutions

Tamper-evident emergency communications using PQC.

- \$3–5B annual flood damage in India
- Compound weather events increasing globally
- PQC migration becoming mandatory for critical infrastructure

References I

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Quantum-Secured Emergency Intelligence Platform

- 5-Qubit VQC cascade prediction
- Dilithium3-secured field communications
- Real + synthetic disaster intelligence
- Unified EOC dashboard

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